

Ximera Euclid's Algorithm, GCD (section 36) COMP 61 (Discrete Math)

Question 1 *The goal of the Euclidean Algorithm is:*

Multiple Choice:

- (a) *Computing $a \bmod b$*
 - (b) *Computing $\gcd(a, b)$ ✓*
 - (c) *Testing if a and b are prime*
 - (d) *Solving $ax + by = 1$*
 - (e) *Solving $ax + by = \gcd(a, b)$*
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Question 2 *The goal of the Extended Euclidean Algorithm is:*

Multiple Choice:

- (a) *Computing $a \bmod b$*
 - (b) *Computing $\gcd(a, b)$*
 - (c) *Testing if a and b are prime*
 - (d) *Solving $ax + by = 1$*
 - (e) *Solving $ax + by = \gcd(a, b)$ ✓*
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Question 3 *If a and b are relatively prime, the smallest positive integer of the form $ax + by$ is:*

Select All Correct Answers:

- (a) *0*
- (b) *1 ✓*
- (c) *2*
- (d) *The GCD of a and b ✓*

Question 4 Which of the following are common divisors of 40 and 56?

Select All Correct Answers:

- (a) -4 ✓
 - (b) 1 ✓
 - (c) 7
 - (d) 8 ✓
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Question 5 What is the gcd of 40 and 56?

Multiple Choice:

- (a) -4
 - (b) 1
 - (c) 7
 - (d) 8 ✓
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Question 6 What is the gcd of 41 and 56?

Multiple Choice:

- (a) 1 ✓
 - (b) 3
 - (c) 7
 - (d) 8
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Question 7 $\gcd(224, 140) =$

Question 8 The key sequence for computing $\gcd(270, 72)$ is: 270, 72, , ,

Question 9 Find integers x and y such that $ax + by = 1$, where $a = 88, b = 45$.

Multiple Choice:

- (a) No solution
 - (b) $x = 5, y = -11$
 - (c) $x = 22, y = -43$ ✓
 - (d) $x = 37, y = -73$
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Question 10 There exists integers x and y such that $ax + by = 1$ for which of the following pair of values for a and b

Multiple Choice:

- (a) $a = 15, b = 105$
 - (b) $a = 3, b = 88$ ✓
 - (c) $a = 6, b = 78$
 - (d) $a = 10, b = 150$
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Question 11 There exists integers x and y such that $ax + by = 1$ _____ a and b are both prime numbers.

Multiple Choice:

- (a) If ✓
 - (b) Only if
 - (c) If and only if
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Question 12 Let a and b be nonzero integers, let $d = \gcd(a, b)$. If e is a common divisor of a and b , then...

Multiple Choice:

- (a) $e|d$ ✓
 - (b) $d|e$
 - (c) $d = e$
 - (d) *None of the above*
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Question 13 What is the gcd of 10 and -5 ?

Multiple Choice:

- (a) 1
 - (b) 5 ✓
 - (c) -5
 - (d) 10
 - (e) -10
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